

sim_1\new\PONG_BASYS3_TB.vhd

```
1  library IEEE;
2  use IEEE.STD_LOGIC_1164.ALL;
3
4
5
6  entity PONG_BASYS3_TB is
7  end PONG_BASYS3_TB;
8
9  architecture PONG_BASYS3_TB_ARCH of PONG_BASYS3_TB is
10
11  constant ACTIVE: std_logic := '1';
12
13  component PONG_BASYS3
14  Port(
15      clk: in std_logic;
16      btnC: in std_logic;    --reset
17      btnR: in std_logic;    --player one
18      btnL: in std_logic;    --player two
19      led: out std_logic_vector(15 downto 0); --game board
20      seg: out std_logic_vector(6 downto 0);  --score display (uncomment for advanced pong)
21      an : out std_logic_vector(3 downto 0) --uncomment for advanced pong
22  );
23  end component;
24
25  signal clk:      std_logic;
26  signal btnC:     std_logic;
27  signal btnR:     std_logic;
28  signal btnL:     std_logic;
29  signal led:      std_logic_vector(15 downto 0);
30  signal seg:      std_logic_vector(6 downto 0);
31  signal an :      std_logic_vector(3 downto 0);
32
33  begin
34
35  process
36  begin
37      clk <= ACTIVE;
38      wait for 5ns;
```

```
39     clk <= not ACTIVE;
40     wait for 5ns;
41 end process;
42
43 UUT: PONG_BASYS3 port map(
44     clk => clk,
45     btnC => btnC,
46     btnR => btnR,
47     btnL => btnL,
48     led => led,
49     seg => seg,    --uncomment for advanced pong
50     an => an       --uncomment for advanced pong
51 );
52
53 PONG_DRIVER: process
54 begin
55
56     --Initialize to Zero
57     btnC <= not ACTIVE;
58     btnL <= not ACTIVE;
59     btnR <= not ACTIVE;
60
61     wait for 1ns;
62
63     --Press reset
64     btnC <= ACTIVE;
65     wait for 1ns;
66     btnC <= not ACTIVE;
67
68     --Wait for when PLAYER_ONE state is set from reset
69     wait until (led(15) = ACTIVE);
70
71     --Start Game (Player1)
72     btnL <= ACTIVE;
73     wait for 100ms;
74     btnL <= not ACTIVE;
75
76     -----
77     -- wait for ball movement
78     wait until (led(0) = ACTIVE);
79     -----
80
```

```
81      --Player 2's turn
82      btnR <= ACTIVE;
83      wait for 100ms;
84      btnR <= not ACTIVE;
85
86      -----
87      -- wait for a miss
88      wait until (led(0) = ACTIVE);
89      -----
90
91      --Player 2's turn
92      btnR <= ACTIVE;
93      wait for 100ms;
94      btnR <= not ACTIVE;
95
96      -----
97      -- wait for ball movement
98      wait until (led(15) = ACTIVE);
99      -----
100
101      --Player 1's turn
102      btnL <= ACTIVE;
103      wait for 100ms;
104      btnL <= not ACTIVE;
105
106      -----
107      -- wait for a miss
108      wait until (led(15) = ACTIVE);
109      -----
110      wait;
111      end process PONG_DRIVER;
112
113      end PONG_BASYS3_TB_ARCH;
114
```